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In vitro and *in situ* abrogation of biofilm formation in *E. coli* by vitamin C through ROS generation, disruption of quorum sensing and exopolysaccharide production

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ABSTRACT

Emergence of antimicrobial drug-resistance amongst food-borne pathogens has led to severe deficit of available therapeutics and requires novel interventions. This study determined the activity of vitamin C (VitC), a natural antioxidant as powerful antibacterial agent against multidrug-resistant (MDR), biofilm-forming *E. coli*. Our findings revealed that VitC wield antibacterial action in dose-time dependent manner with minimum inhibitory concentration (MIC) of 125 mM. At these concentrations VitC impaired quorum sensing (QS) and exopolysaccharide (EPS) production and induced sugar and protein leakage from the bacterial cells by virtue of reactive oxygen species (ROS) generation. Furthermore, VitC-treated bacteria showed downregulation of genes underpinning biofilm signaling (*luxS*) and regulation (*bssR*) by up to 27-folds. Finally, this study demonstrated the promising antimicrobial application of VitC, *in situ,* in Indian soft cheese (paneer) when applied as a coating. Therefore, VitC can be applied as natural and safe 'antimicrobial' against biofilm-forming bacteria in food systems vis-à-vis other conventional antimicrobials.

1. Introduction

Antibiotic resistance is a massive threat and is presently growing as a global menace. Most of the powerful antibiotics are losing their efficacy at an uncertain pace due to the rapid evolution of bacterial defences, genetic exchange and their multi-drug tolerance. Although antibiotics are known to be quite effective against planktonics (free-living cells), conversely the biofilms community of microorganisms, which adhere to a surface by arranging in a three-dimensional structure (Helgadóttir et al., 2017) display 10–1000 fold resistance. Bacteria present in biofilms portray multifactorial mechanisms wherein the penetration of antibiotics inside the biofilm structure is hindered which ultimately deactivate the functioning of antibiotics. The spread of antimicrobial resistance (AMR) and MDR pathogens through the food supply chain poses a very serious and insurmountable food safety threat which has eventually led to a huge public health challenge. Each year in U.S. at least 2 million people suffer from antibiotic resistance infection and more than 23,000 people die (http://www.cdc.gov/drugresistance/index.html). The infections, which were known to be trivial, are becoming untreatable due to the emergence of new MDR strains of bacteria and have resulted in significant morbidity and mortality in the healthcare.

In order to sustain good health in present era, the access to healthy and nutritious food is crucial. However, with the advent of 21st century, with improved sanitary, hygiene practices and advanced methods of food preservation and diagnostics; it is ironic that the basic food safety is still posing a great challenge. Hitherto reports have indicated that over 600 million people become ill after consuming contaminated food and that 420,000 die each year (Havelaar et al., 2015).

Numerous outbreaks related to *E. coli* have been a leading cause of food safety threat to the human beings. In most of the African, Asian and Latin American countries *E. coli* is the leading cause of acute diarrhoea infection in children < 5 year old (Imdad et al., 2018). Shiga toxin-producing *Escherichia coli* (STEC) are an important cause of illness in United States (https://www.cdc.gov/ecoli/2019/bison-07–19/index.html).

Biofilm-forming ability of some *E. coli* strains has been considered to be an important virulence factor used by pathogens to overcome host

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immune system, antimicrobial stresses and transmission (Safadi et al., 2012). Biofilms are responsible for various human infectious diseases, industrial corrosion and food contamination (Flemming et al., 2016). Within the biofilm, bacteria interact with each other and produce dense exopolysaccharide (EPS) which in turn provides the bacterial community protection from hostile environment and exposure of antibiotics and antimicrobial agents (Høiby et al., 2010; Wu et al., 2013). The extracellular matrix lowers the penetration of antibiotics and antimicrobial agents there by reducing the effective dose to be delivered into a cell (Helgadóttir et al., 2017).

Failure of existing antibiotics to eliminate biofilm-associated bacteria, necessitate multiple and intensive antimicrobial regimes that further enforce the development of drug-resistant pathogens and exhaustion of last resort antimicrobials. Hence there is an urgent need for safer alternatives especially which are natural, non-toxic and do not generate resistance to the pathogens. VitC (Ascorbic Acid), a natural antioxidant compound is one such alternative which has no adverse effect, is cheap and easily available. Most importantly it has been used successfully to treat health ailments since it plays a vital role in the maintenance of collagen, act as a cofactor in the conversion of dopamine to norepinephrine and is involved in the synthesis of catecholamine. Conspicuously, VitC catalyzes enzymatic reaction for the activity of certain hormones, which helps in mitigating oxidative injury and restores microcirculatory flow, boosts antimicrobial defence (Verghese et al., 2017). Additionally, the food industries are interested to develop some specific additive of VitC which can be used in different food products (Varvara et al., 2016). The strong antioxidant activity of VitC and its ability to establish the link with other essential nutritional factors has made VitC an additive leading to various transformations in food processing industries (Bauernfeind & Pinkert, 1970).

Food processing industry resist choosing chemical and thermal methods for retarding microbial growth since it affects organoleptic and nutritional quality of food and also causes negative health impact such as allergic reaction due to nitrates, benzoates, sulfites etc (Quinto et al., 2019). Reports have also suggested that VitC based additives when employed alone or in combination with antimicrobial agents inhibit synthesis of EPS and quorum sensing and induces oxidative stress, resulting in killing of gram-positive bacterium, *Bacillus subtilis* (Hall and Mah, 2017; Pandit et al., 2017; Verghese et al., 2017).

Therefore, in preview of the above mentioned important role of VitC, the present study evaluates the anti-biofilm and antimicrobial efficacy of VitC against a strong biofilm forming, multidrug resistant *E. coli* EMC17 isolate. Herewith the results demonstrated its efficacy in preventing microbial growth *in vitro* as well as *in situ* within perishable food matrix (Indian cottage cheese).

2. Materials and methods

2.1. Bacterial strain and growth conditions

E. coli EMC17, laboratory isolate from meat sample, was used for the current study (a detailed methodology is given in the Supplementary material). The bacterial culture was grown in TSB (Tryptic Soy Broth, HiMedia) at 37 °C/24 h/180 rpm under shaking conditions and on solid Nutrient agar plates (HiMedia) at 37 °C/24 h under static conditions. A standard OD_{600} (~0.1) corresponding to ~1.0×10^6 Colony Forming Units (CFU)/mL was applied throughout the experiment with or without dilution. All the media and chemicals were purchased from HiMedia or Sigma-Aldrich unless mentioned otherwise.

2.2. Biofilm formation

The biofilm forming ability of EMC 17 was evaluated as previously described by Champion, Catanzaro, Bandara, & Inzana, (2019) with slight modification. Briefly, overnight grown culture (Stationary phase) at 37 °C were diluted to 0.1 OD. The culture was incubated for 24, 48, 72 and 96 h on a 96 well polystyrene plate. The medium was replaced every day. The mature biofilm formed was washed twice with sterile 1X PBS and were stained with Crystal Violet (CV) for 30 min. The attached biofilm was quantified by solubilizing the CV with 95% ethanol and absorbance was noted at 540 nm in an ELISA reader (Infinite F50, Tecan, Switzerland).

2.3. Minimum inhibitory concentration (MIC) of VitC

The method to determine the minimum inhibitory concentration of VitC was taken from the studies of Taneja et al., (2013). A two-fold serial dilution of VitC (L-Ascorbic acid, Sigma) stock was prepared in fresh NB (Nutrient Broth, HiMedia) and freshly grown culture (log phase) with a cell density of ~1.0×10^6 CFU/mL were added onto a 96 well microtitre plate (polystyrene) which provided a surface for adherence for biofilm development. For the untreated control wells, only culture and media were added. The plate was incubated at 37 °C for 24 h, under static conditions. After the incubation period, the supernatant containing planktonic lifestyles were discarded by removing the media from each well, followed by washing twice with sterile 1X Phosphate Buffer Saline (PBS). The adhered cells were recovered in 100 µL PBS for the serial dilution and plating on nutrient agar plates was performed subsequently. Percent inhibition of biofilm formation was determined by dividing the number of CFU obtained after VitC treatment by the number of CFU obtained from untreated controls. Each experiment was performed in triplicates. MIC was considered as the lowest concentration of VitC that permitted the growth of ≤1% of the bacteria in comparison to the observed in the untreated controls (Taneja & Tyagi, 2007).

2.4. Quorum sensing

Evaluation of quorum sensing was done according to the method described by Kawaguchi, Yung, Norman, & Decho, (2008) with slight modification. A setup of freshly grown cultures (1.0×10^6 CFU/mL) with VitC (125 mM) was incubated at 37 °C at different time points: 4, 6 and 8 h under static condition respectively. After each time point, cells were harvested by centrifugation (10000 rpm, 15 min). The supernatant was mixed with ethyl acetate in 2:1 ratio followed by vortexing for 10 min. This mixture was then incubated at room temperature for the separation of aqueous and organic components. Upper layer, potentially containing the acyl homoserine lactones, was transferred into sterile falcon and incubated at 40 °C for 15 mins for further analysis. Thereafter 50 µL of 1:1 mixture of 2 M hydroxylamine and 3.5 M NaOH was added to 40 µL of test sample. Further 50 µL of 1:1 mixture of ferric chloride solution (10% in 4 M HCl) and 95% ethanol was also added to the previous mixture made. Quorum sensing activity was measured spectrophotometrically at 540 nm (SICAN UV–VIS Spectrophotometer, Inkarp, India).

2.5. EPS production analysis

EPS was extracted according to the procedure described by Jung, Choi, & Lee, (2013) with slight modifications. An overnight grown culture was harvested by centrifugation (15000*g* for 5 min) and boiled for 15 min at 100 °C. Thereafter, 20 µL Proteinase K (Sigma Aldrich) was added after 20 min and further incubated at room temperature for an hour. Subsequently, 100 µL of 85% Trichloroacetic acid (TCA) was added to the sample and the tubes were further incubated for 30 min on ice followed by centrifugation at 15,000g for 20 min. Supernatant was collected and an equal volume of 95% ethanol was added. The mixture was then centrifuged (15,000*g* for 20 min) after an incubation at 20 °C for 1 h. The precipitate was collected and then washed twice with 95% ethanol and centrifuged at 15,000g for 20 min. Final precipitate was dissolved in 1 ml of deionized distilled water and stored at −20 °C. Total EPS (expressed as mg per microliter) was estimated by the phenol–sulfuric method (Dubois et al., 1956) and analysed spectrophotometrically at 490 nm (SICAN UV–VIS Spectrophotometer, Inkarp, India).

2.6. Epifluorescence microscopy

The Epifluorescence microscopic study was performed as described by Bogachev et al., (2018). Briefly, borosil glass slides were made sterile by soaking in ethanol (70%) for 10 min and placed in a sterile 50 ml falcon containing the test sample and the control. The falcons were incubated at 37 °C for 24 h allowing the biofilm to develop. After incubation, slides were removed from the growth medium and washed twice in sterile 1X PBS to remove any loosely attached cells. The cells were fixed onto the slide with the help of 2.5% glutaraldehyde for 30 min. The slides with the fixed cells were stained in Fluorescent Di-Acetate (FDA) for 6 min in dark (Taneja et al., 2010).

Further, the slides were washed twice with sterile 1X PBS to rinse off the residual stain, followed by air-drying of the slides and microscopy. Biofilms formed on the glass slides were viewed under the fluorescence microscope (Nikon Eclipse ci microscope, Japan) and images were captured.

2.7. Time kill-kinetics study

VitC was tested to determine the time-kill kinetics as described by Appiah, Boakye, & Agyare, (2017) with little modifications. Briefly, logarithmically grown culture (~1.0×10^6 CFU/mL) was inoculated with VitC (125 mM) and the plate was incubated at 37 °C under static condition. Serial dilution was performed at each time point (1/2,2,4,8,16 and 24 h) as described in the section 2.3. A graph of log CFU versus time was plotted. Each experiment was performed in triplicates.

2.8. ROS determination

To determine pro-oxidant role of VitC against test bacterium and evaluation of endogenous ROS production, a DCFDA (2′,7′ –dichlorofluorescein diacetate)-based assay was performed as described by Qayyum, Oves, & Khan, (2017) with slight modification. DCFDA is a colourless, non-fluorescent dye which passively diffuses into the cell where it is cleaved by non-specific intracellular esterases. The cleaved product gets easily oxidized by intracellular ROS, yielding highly fluorescent compound, DCF (2′,7′ –dichlorofluorescein). Overnight culture of *E. coli* EMC17 was taken and the Optical Density (OD_{600}) was set to 0.1 in 10 ml NB. A freshly prepared VitC solution was added at a final concentration of 125 mM. The tubes were then incubated at 37 °C for 4 h, followed by centrifugation at 8000 g at 4 °C for 10 min. The supernatant obtained was collected separately and incubated with 10 μM DCFDA dye for 1 h at 37 °C. Fluorescence was measured by excitation at 485 nm and emission at 520 nm using Cary Eclipse fluorescence spectrophotometer (Agilent, USA). A similar kind of experiment was set up for culture-free blank as well.

2.9. Reducing sugar and protein leakage analysis

In order to determine the VitC induced damage and subsequent membrane disruption, analysis for leakage of reducing sugar and proteins from treated bacterial cells were determined as described by Yuan, Peng, et al., (2017). Rupturing of bacterial cell post-VitC treatment would lead to release of intracellular biomolecules like reducing sugars and protein in the extracellular milieu. The increase in the biomolecular concentration following treatment is a strong indicator towards bacterial membrane damage. The test cultures of a cell density ~1.0×10^6 CFU/ml were treated with VitC (125 mM) for 4 h. The treated cells were centrifuged at12000 rpm for 30 min and the supernatant was collected for further analysis. The concentration of reducing sugars was estimated using Dinitro salicylic acid (DNSA) assay whilst the protein was estimated by Bradford assay. The sugar content was determined using a glucose standard curve and expressed as μg/μl of glucose and protein content was expressed as μg/200 μL of Bovine Serum Albumin (BSA).

2.10. Gene expression

2.10.1. RNA isolation

The petriplates containing test culture treated with sub-inhibitory concentration of VitC (MIC_{50}) and untreated control plate were incubated at 37 °C statically for 24 h. The treated and untreated culture cells were pelleted down for bacterial RNA extraction using RNeasy mini kit (Qiagen Technologies, Hilden, Germany) according to manufacturer protocol.

2.10.2. cDNA synthesis

The High-quality total RNA was further used for the synthesis of cDNA using high-capacity cDNA reverse transcription kit (Applied Biosystems) using manufacturers protocol.

The RNA and cDNA were quantified spectrophotometrically using NanoDrop-One (Thermo Scientific).

2.10.3. Quantitative RT-PCR (qRT-PCR) analysis

qRT-PCRs were performed for various gene targets using the Power UP^{TM} SYBR Green master mix (Applied Biosystem, Austin) on CFX 96 Touch Real-Time PCR detecting System (Bio-Rad) as per the manufacturer's recommendations. Amplification specificity was confirmed by melt curve analysis. The gene expression levels were analysed using livak method after normalization using housekeeping gene *rrsH* and expressed as fold-difference in expression (VitC treated vs untreated control biofilms). All the primers used in this study are enlisted in Supplementary Table S1.

2.11. Monitoring E. coli growth and survival in Indian cottage cheese (Paneer)

The paneer for the experiment was prepared as described by Sharma, Shivaprasad, Chauhan, & Taneja, (2019). Uniformly cut slices (1 g) of paneer were taken and subjected to dip-coating in a solution of VitC (125 mM concentration) for 2 min and air-dried. This was repeated 2–3 times. Uncoated paneer samples were taken as control. Thereafter, logarithmically grown *E. coli* EMC17 culture (OD_{600} ~ 0.1) was harvested by centrifugation (5000 rpm for 10 min) and spiked on paneer by dipping them in 0.001 OD (~$5x10^5$ CFU) culture for 2 min. The paneer slices were air-dried for 10 min. The VitC-treated and untreated paneer samples were vacuum-packed and stored at 4 °C. The samples were drawn aseptically at various intervals of time (1 to 10 days) and were serially diluted (10-fold) before plating them on Eosin Methylene Blue (EMB) agar plates (triplicates). The plates were incubated at 37 °C for 24–48 h and bacterial count was taken to enumerate the survivors. The schematic overview of the experimental set-up followed is shown in Fig. 4a.

2.12. Sensory analysis

Sensory evaluation of prepared paneer was performed on a 9-point hedonic scale by a panel of 10 members with corresponding descriptive terms ranging from 9 'like extremely' to 1 'dislike extremely'. The mean score of the sensory attributes were represented in a spider chart (Raju & Sasikala, 2016; Sharma et al., 2019).

2.13. Statistical analysis

All the experiments were performed in triplicate. The results are represented as mean ± standard deviation. The statistical significance of the results (VitC-treated versus untreated control sets) has been

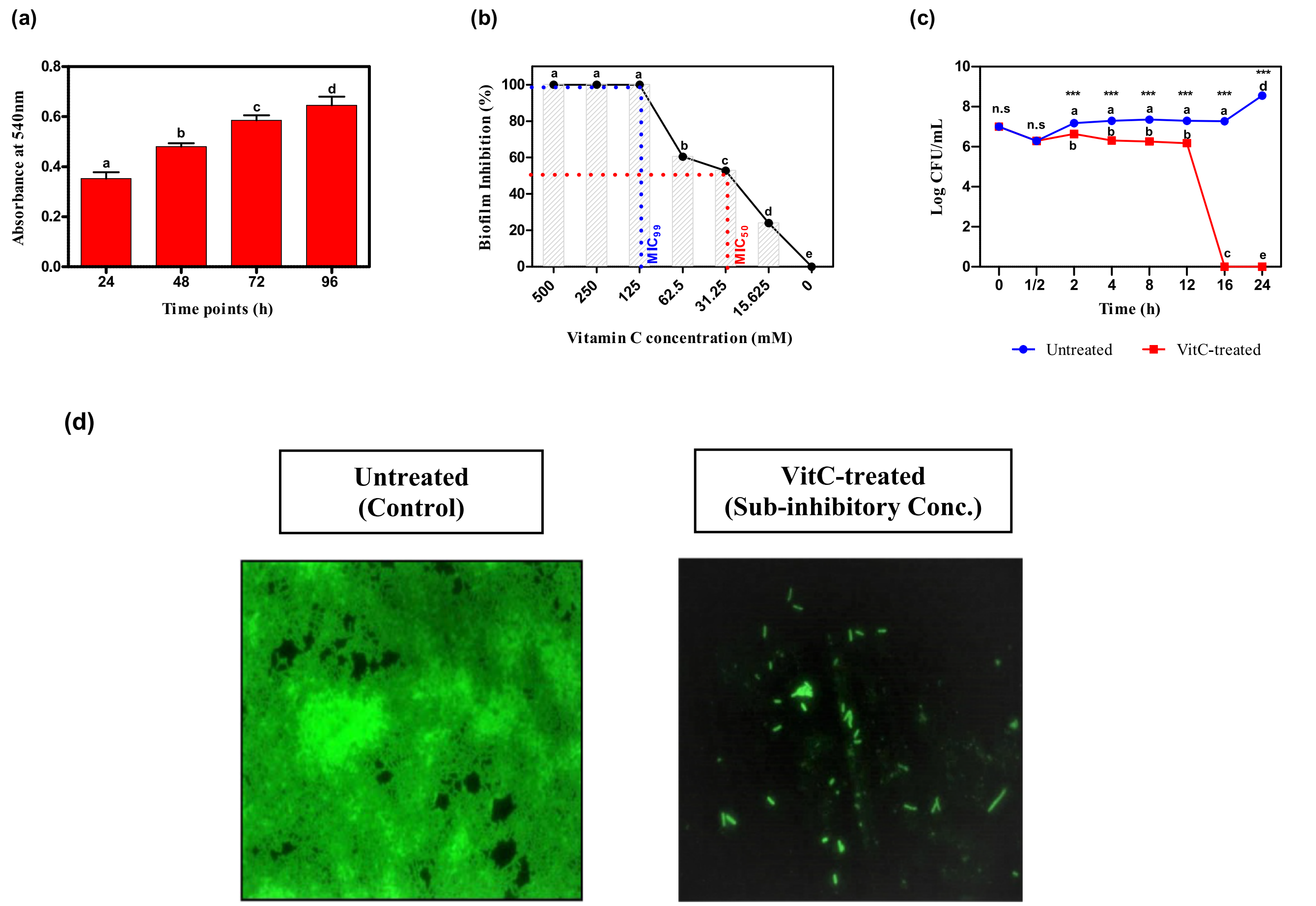

Fig. 1. Biofilm formation of *E. coli* EMC17 and vitamin C mediated disruption of biofilms. (a) Time kinetics of biofilm formation (b) Effect of different concentrations of vitamin C (c) Time kill kinetics of vitamin C (125 mM concentration) (d) Fluorescence microscopic images of biofilms on surface of glass slides [untreated (control) vs Vitamin C-treated (MIC_{50} dose) culture]. Mean ± SD from 3 replicates plotted for all panels; Means with different letters are significantly different (p- value *** < 0.001, ** < 0.01, * < 0.05, ns, not significant).

evaluated using unpaired two tailed *t*-test. $P < 0.05$ was considered to be statistically significant.

3. Results and discussion

3.1. Evaluation of biofilm forming ability of E. coli EMC17 isolate

Biofilm formation is an important characteristic of *E. coli* and other enterobacteria, often serving as an important virulence factor (Ramos-Vivas et al., 2019). Therefore to evaluate the biofilm forming ability, *E. coli* EMC17 isolate was assessed for biofilm formation on polystyrene surface using Crystal Violet assay at different time points (24, 48, 72 and 96 h). The study revealed that EMC17 isolate possess a high capacity of biofilm formation on plastic surfaces such as polystyrene and maximum biofilm formation occurred within 96 h of incubation on a microtiter plate (Fig. 1a). Detailed characterization of the *E. coli* EMC17 (meat isolate) in our laboratory has established it as a MDR, biofilm-forming strain belonging to extra-intestinal pathogenic *E. coli* (ExPEC) (Bharadwaj et al., 2020). Previous studies have demonstrated that pathogenic *E. coli* strains have the ability to attach to different surfaces, including food processing surfaces and form biofilm (Cookson et al., 2002; Ryu et al., 2004; Vidal et al., 1998).

3.2. Antimicrobial activity of VitC

VitC is a pleiotropic molecule possessing dual characteristic of an antioxidant (prevents oxidation of molecules) and a pro-oxidant (promotes oxidation) molecule which is primarily governed by its sub-cellular concentration as well as environmental conditions (Vilchèze et al., 2013). The antimicrobial activity of VitC was examined against the bacterial biofilms formed by *E. coli* EMC17 isolate. This is the first insight wherein we evaluated the effect of VitC against planktonic and biofilm-state of *E. coli* EMC 17. Notably, we found 125 mM concentration (Fig. 1b) as a MIC of VitC that inhibited the *E. coli* biofilms, while the lower concentrations of VitC constrained the growth of biofilm bacteria in a concentration-dependent manner with 50% killing (MIC_{50}) observed at 30 mM concentration. The observed MIC corroborates well with the previous findings by (Zhang et al., (1997) that depict the MIC of VitC for gram negative bacteria (*Helicobacter pylori*) ranges between 2.048 mg/mL (11.62 mM) & 16.384 mg/mL. (93.027 mM). Likewise, in another study Sendamangalam, (2010) reported 2 mg/mL (11.355 mM) as MIC of VitC against *Streptococcus mutans* which reduces the biofilm formation by 75.7%. To our surprise we found that VitC completely inhibited the growth of planktonic cultures of *E. coli* EMC17 (Supplementary Fig. S1) at a lower concentration ($MIC_{99} \sim 125$ mM and $MIC_{50} \sim 15$ mM). This is in contrast to the results reported by Pandit et al. (2017) wherein they showed that VitC, at low concentrations did not exhibit any bactericidal effect against planktonic bacteria. Whilst, they predicted the capacity of VitC to destabilize the biofilms of *Bacillus subtilis* by inhibiting production of EPS matrix, reducing the biofilm complexity and enhancing the drug susceptibility to various antimicrobial agents. In view of the variant studies in *Mycobacterium tuberculosis*, a fairly high concentration of VitC was found to be essential for an efficient antibacterial affect against

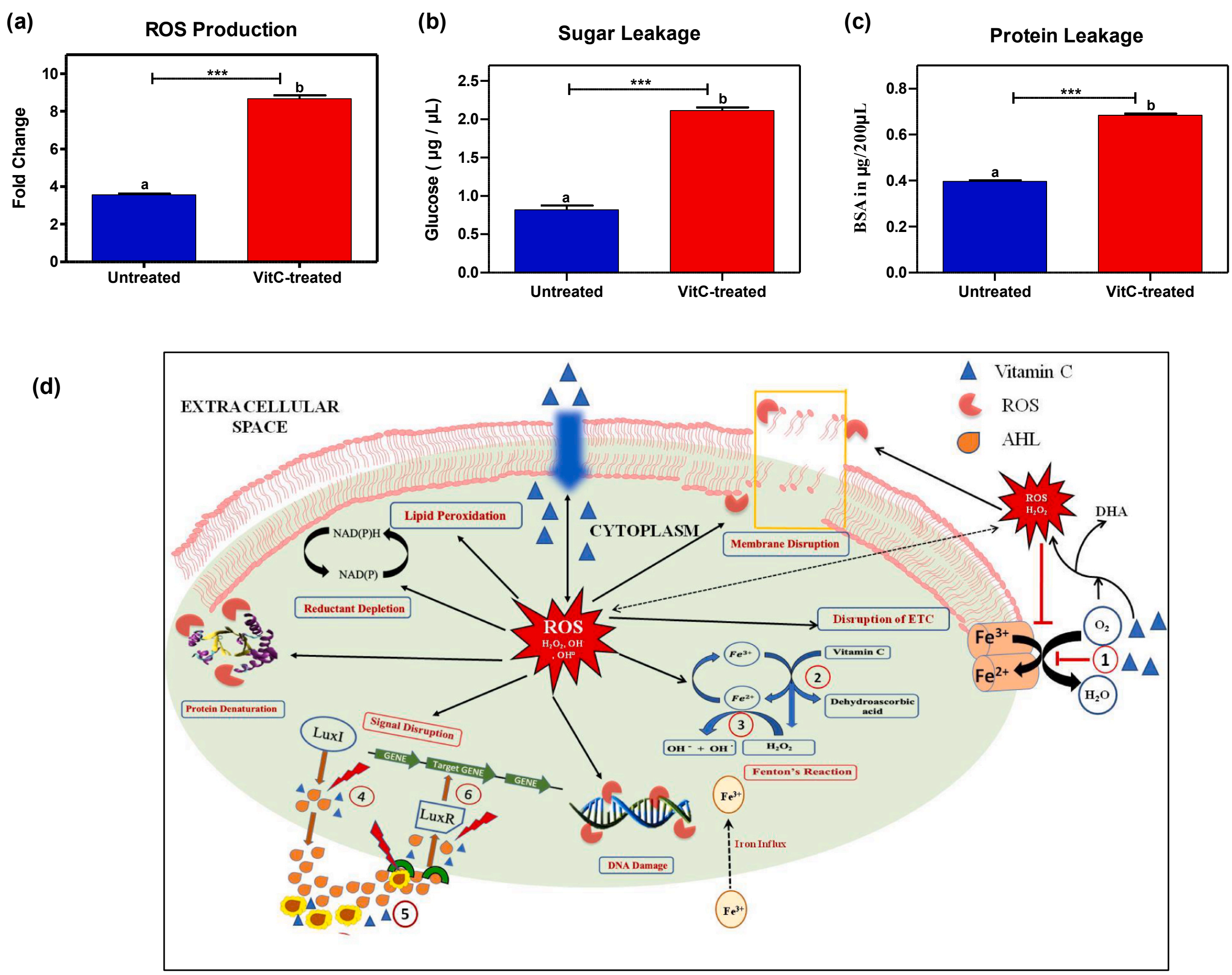

Fig. 2. Bactericidal mechanism of pro-oxidant concentrations of vitamin C (at MIC dose-125 mM). (a) ROS detection using DCFDA assay; (b) Sugar leakage measured using DNSA assay and (c) Protein leakage measured using Bradford assay for *E. coli* EMC17 culture treated with lethal dose of vitamin C for 4–12 h vs. untreated control; (d) Schematic diagram depicting bactericidal mechanism of vitamin C due to its pro-oxidant (ROS generating) activity. Extracellular ROS generated by reaction between O_2 and vitamin C thereby generating H_2O_2 which induces extracellular membrane damage as well as disruption of the electron transport chain (reaction 1); Intracellular vitamin C induces Fenton's reactions (2 and 3) and generation of ROS (OH° and OH^-) storm that induces sub-cellular damage: DNA damage, protein denaturation, lipid peroxidation and Fe-S cluster disruption leading to disruption of the electron transport chain, depletion of reducing power of cell [NAD(P)H] and inhibition of quorum sensing pathway by: lowering production of quorum sensing molecules AHL's 4; vitamin C directly interacts with AHL's 5 and blocks signal reception which further effects all downstream signal transduction pathways linked to biofilm formation 6 and results in disruption of biofilms. Mean ± SD from 3 replicates plotted for all panels; Means with different letters are significantly different (p- value *** < 0.001).

actively growing cultures (Vilchèze et al., 2013). Additionally few other studies suggest that VitC enhances the susceptibility of *E. coli* towards antibiotics and cold plasma treatments, without being directly involved in reducing the viability or demonstrating the bactericidal activity (Kallio et al., 2012; Khameneh et al., 2016; Pandit et al., 2017).

With respect to the above-mentioned reports, the observed differences of low or no bactericidal activity of VitC is probably attributed to the variations in the methodologies adopted during the study i.e. differences in the bacterial culture used, test cell density, media composition and test concentrations of VitC. Looking at the in-depth of variations from the present study, a food (meat) isolate *E. coli* EMC17 was used, which is a strong biofilm former belonging to ExPEC family of *E. coli* that are well credited as biofilm formers. Further, the current study employed TSB medium for bacterial growth against the LBGM medium (LB medium containing 1% glycerol, 1 mM MnSO4) used by Pandit et al. (2017). The LBGM medium contains Manganese, whose presence in the media hinders the availability of VitC during the experimental course of action (Supplementary Fig. S2), which otherwise as mentioned in the mechanism (Fig. 2d), is consumed by Fe^{3+} present in the bacterial membrane that leads to generation of ROS (via Fenton's reaction) and ultimately damages DNA and kills bacteria (Fig. 2d). This can be explained by the known fact that Mn^{2+} has higher tendency to form complex with VitC than Fe^{2+} since Mn^{2+} has small size and high charge density than Fe^{3+} (Supplementary Fig. S2), and therefore the competition arises which leads to formation of more stable manganese ascorbate than reacting with Fe^{3+} to generate ROS which would have otherwise elicited the bactericidal effect. The test concentration of VitC used by Pandit et al. (2017) ranged from 10 to 40 mM, which effectively did not reach the minimum threshold for generating ROS via VitC (ROS is key to its bactericidal mechanism) due to presence of Mn^{2+} in the media. Furthermore, it is well known that when testing antimicrobial activity of compounds, cell density influences the resultant MICs. High cell density reduces the efficacy of antimicrobials and requires a higher dose of killing the microbial cells (CLSI 2010). Moreover, the authors of the latter study used high cell density to determine bactericidal activity of VitC against both planktonic ($1x10^7$ CFU/ml) and biofilm (5×10^6 CFU/ml) bacteria.

3.3. Time kill kinetics

The time kill kinetics was obtained by treating EMC17 with lethal concentrations of VitC (125 mM). The bactericidal activity of VitC began early on and completely inhibited *E. coli* biofilms within 16 h of treatment (Fig. 1c). The time-kill studies are often used to determine the effect of antimicrobials against bacteria over a period of time (Lemos et al., 2018). When the concentration of antibacterial compound surpasses the MIC value, time-dependent bactericidal effect takes place (Anantharaman et al., 2010; Lemos et al., 2018). The present study depicted the slow and steady inhibition of biofilm because of gradual build of ROS moieties (section 3.5 below) which generates an oxidative stress and disrupts the bacterial quorum sensing, EPS production and damages the bacterial cell membranes bringing about cell death and biofilm disruption.

3.4. Epifluorescence microscopic study

Having established the anti-biofilm effect of VitC against EMC17, we investigated the impact on biofilms formed on the glass slides under the fluorescence microscope (Nikon Japan). The cultures grown in presence (MIC_{50}, sub-lethal dose) or absence of VitC, followed by FDA staining were visualized via Fluorescence microscopy; the images showed the extent of biofilm disruption and scarcity of bacterial biomass in VitC-exposed cultures vis-à-vis the biomass abundant control group (Fig. 1d). The actual mechanism of biofilm reduction by VitC has been discussed later in section 3.5. It is evidently shown that the pro-oxidant property of VitC (at higher concentrations) leads to ROS generation, which results in bacterial killing and reduced cellular growth in the presence of VitC.

3.5. Bactericidal mechanism of VitC

3.5.1. ROS generation

The pro-oxidant action and subsequent ROS generation by VitC has been implicated as the foundation for its antibacterial activity. The reduction of ferric ions, (Fe^{3+}) to ferrous ion (Fe^{2+}) is induced by excess levels of VitC, which further leads to excess intracellular load of ferrous ions (Amatore et al., 2008; Guaiquil et al., 2001; Vilchèze et al., 2013; Zhong et al., 2017). The excess ferrous ions lead to the generation of ROS species (hydroxyl radicals, hydrogen peroxide and superoxide) via Fenton's reaction (Vilchèze et al., 2013), which thereby damage various intracellular targets and bring about cell death (Fig. 2d, Supplementary text S1). To evaluate the role of VitC in inducing ROS-mediated killing of the EMC17 biofilms, intracellular ROS accumulation was measured using the DCFDA assay. The assay depicts the fluorescence generated which is ultimately linked to the quantification of the ROS generation in the cell. In comparison to the untreated control, VitC treatment induced an 8.541 folds increase in fluorescence in EMC17 biofilms (Fig. 2a). The rise in the fluorescence intensity confirmed the increased production of ROS in the cells. Furthermore, the ROS production decreased with decreasing concentration of VitC indicating that pro-oxidant role of VitC was concentration dependent. This corroborates well with the inhibition of biofilms we observed at varying VitC concentrations (Fig. 1b). Oxidative stress is an important part of host innate immune response to foreign pathogens and pathogens are eliminated intracellularly (Chen et al., 2014). Previous studies on *Mycobacterium tuberculosis* have reported that VitC produces reactive oxygen species and exhilarate lipid peroxidation leading to bacterial inhibition (Vilchèze et al., 2013; Yuan, Peng, & Gurunathan, 2017). The common mechanism behind bactericidal action of VitC is the production of reactive oxygen species by Fenton's reaction (Ghosh et al., 2019; Sikri et al., 2018; Taneja et al., 2010; Vilchèze et al., 2013). The increase in the pro-oxidant activity of the VitC results in the influx of the iron inside the cell resulting in generation of ROS via Fenton's reaction. It has been well documented and confirmed that, free radical generation is involved in membrane disintegration, DNA damage, disruption of the iron-sulfur cluster, the influx of iron, and disrupting biosynthesis pathways (TCA and other metabolic pathways) ultimately resulting in killing of microbes such as *E. coli, S. aureus, S. epidermis, P. aeruginosa* etc. (Acker & Coenye, 2016; Ghosh et al., 2019) and is schematically depicted in Fig. 2d.

3.5.2. Protein and sugar leakage

ROS generation-induced by VitC that ultimately led to bactericidal effect in EMC 17 also promoted cell membrane destabilization and disruption, leading to the leakage of intracellular biomolecules like proteins and reducing sugars. The evaluation was done by assessing for leakage of intracellular biomolecules like proteins and reducing sugars post-VitC exposure of EMC 17. As shown in Fig. 2b and 2c, the leakage assay reveled 2.5 folds increase in the amount of reducing sugar and 1.4 folds increase in the amount of proteins in the extracellular milieu after 4 h of VitC treatment vis-à-vis untreated control samples. These results confirm membrane disruption and subsequently bacterial lysis upon VitC exposure. Natural antimicrobials have been implicated in rupturing cell membrane, affect nucleic acids stability and actions, decay proton motive force, deplete adenosine triphospahate (ATP) etc., (Quinto et al., 2019).

3.6. Effect of VitC on quorum sensing (QS) and EPS production

Quorum sensing is a bacterial cell–cell communication in which small chemical molecules (called auto-inducers (AI's) like acyl homoserine lactones (AHL's) for gram-negative bacteria) are produced and sensed leading to a cascade of reactions which regulate multiple genes responsible for expression of virulence factors, biofilm formation, entry to stationary phase, spore formation etc. (Jiang et al., 2019). Fig. 3a demonstrates that VitC treated culture exhibited ~1.5 fold reduction in quorum sensing activity compared to their untreated control over different intervals of time (4, 6, and 8 h). We believe that VitC would be affecting one or more of the three major steps which determine the QS activity: (i) Inhibits the luxI-like genes involved in production of QS signal, (ii) sequesters the effector molecules (AHL's) or blocks the surface receptors for QS molecules leading to signal scrambling and/or (iii) interferes with signal reception, transmission and down-regulation of genes responsible for biofilm-formation. As per our findings detailed in this section and section 3.7, we have shown that VitC treatment abrogates the QS signals, EPS production and down-regulates the gene expression of *luxS* and *bssR* genes responsible for the production of QS effector molecules (HSL, homoserine lactone), which probably is further disintegrated in the presence of the free radicals ($OH^{\cdot}$ and OH^{-} generated via VitC) thereby leading to alkaline hydrolysis (Gómez-Bombarelli et al., 2013) of the effector molecule and signal transduction respectively, thus ensuing obliteration of biofilm formation (Supplementary Fig. S3).

Since QS step majorly regulates the subsequent steps of biofilm formation (including EPS production, microcolony formation, dispersal), it was logical to believe that EPS production post VitC treatment would be compromised in the treated cells. The principal functions of EPS include the initial attachment of cells on various surfaces and safeguarding against environmental stress and dehydration (Vu et al., 2009). As shown in the Fig. 3b, the VitC-treated bacterial culture showed a 5.8 folds reduction in EPS production (0.5 μg/μL) in comparison to control (2.9 μg/μL). This indicates that VitC hampers the production of EPS in *E. coli* EMC17. Pandit et al. (2017) have previously demonstrated the anti-biofilm effect of VitC against *B. cereus* biofilms its impact on reduction of quorum sensing and EPS secretion at sub-lethal concentration. Since EPS is a complex mixture of biopolymers consisting of polysaccharides as well as protein, nucleic acids, lipids and humic substances (Vu et al., 2009), its absence or deficit would result in more exposed cells that would be susceptible to most bioactive compounds and conditions. The same can be corroborated from the

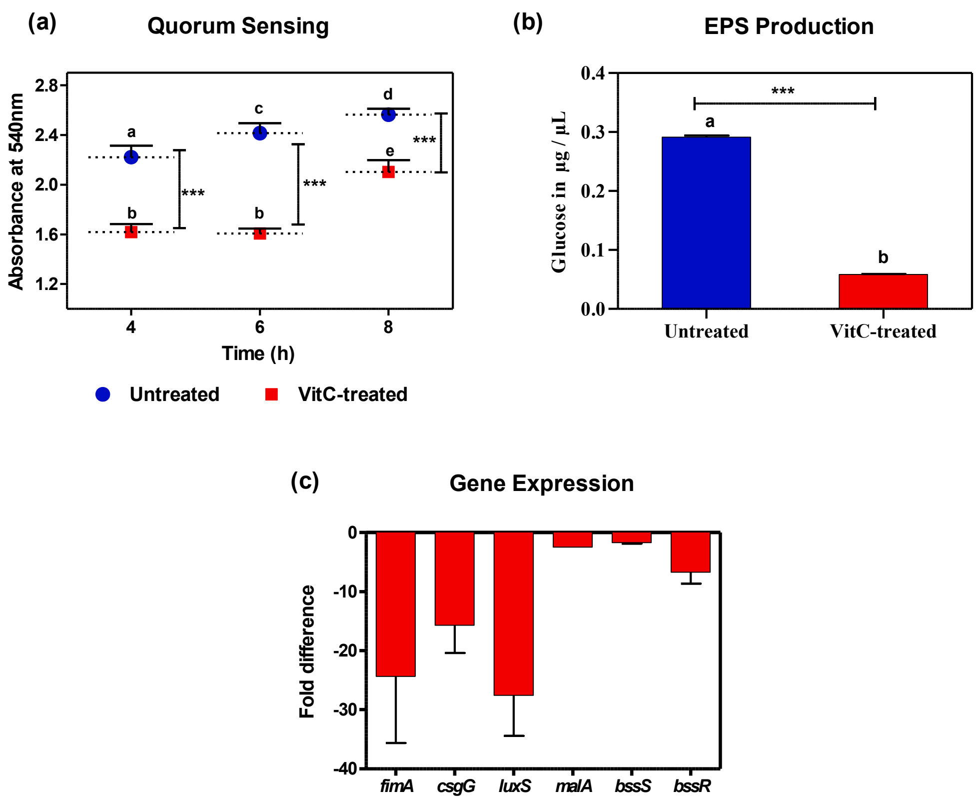

Fig. 3. Effect of vitamin C (vit C) on *E. coli* EMC17 (a) Quorum Sensing activity (b) Exopolysaccharide (EPS) production and (c) Fold expression of biofilm formation genes in *E. coli* EMC17 cells treated with or without vitamin C (at MIC_{50} concentration). Mean ± SD from 3 replicates plotted for all panels; Means with different letters are significantly different (p- value *** < 0.001).

fluorescence microscopic studies discussed earlier in section 3.4 above.

3.7. Gene expression

Further to support the anti-biofilm activity of VitC and the underlying bacterial killing, a transcriptional response of EMC17 was examined in the presence of VitC. The results showed that expression levels of all the tested genes imperative/vital to *E. coli* biofilm formation (*fimA, csgA, luxS, malA* and *bssR*) were down-regulated (Fig. 3c). Therefore, it was speculated that antibiofilm activity of VitC was because of the inhibition in the transcription level of biofilm forming genes.

To further confirm the inhibitory effect of VitC on biofilm formation, the transcript levels of biofilm-associated genes of EMC17 (*fimA, csgG, luxS, malA, bssS* and *bssR*,) were tested by performing real-time RT-PCR experiments (Bio-Rad CFX 96 Touch Real-Time PCR detecting System). Result confirmed that there was a decrease in transcription of biofilm-associated genes *fimA, csgG, luxS, malA*, and *bssR* compared to untreated culture, thus confirming the anti-biofilm activity of VitC against EMC17.

The gene *fimA*, encodes for major type I subunit fimbrin (pilin) in *E. coli,* which is important for the biofilm formation (Niba et al., 2007). While *csgA* encodes for curli formation in *E. coli* (Hammar et al., 1995). Compared to control (untreated), the expression of the *fimA* and *csgA* were highly downregulated with *fimA* showing 24-folds and *csgA* showing 15-fold reduction in expression. The *luxS* gene encodes for the production of AI-2 (Autoinducer 2), a bacterial QS molecule which helps in cell to cell communication (Schauder et al., 2001; Yang et al., 2005).

We observed a 27-fold decrease in the expression of the *luxS* gene under VitC-treated biofilm condition. The decreased expression of *luxS* gene corroborated our findings of decreased production of QS and AHL molecules upon treatment with VitC consequently affecting the biofilm formation.

The enzyme MalA encoded by *malA* gene, helps in the easy uptake and utilization of maltose or maltodetxtrin by hydrolyzing them to glucose. The expression of this gene was downregulated by 2 folds compared to the control. The genes *bssS* and *bssR* encodes for regulation of biofilm through signal secretion (Da Re et al., 2013; Domka, Lee, & Wood, 2006). The study observed that the expression of *bssR* gene was downregulated by 6 folds, while *bssS* was not affected, as compared to control. Previous study of VitC treated biofilms of *B. subtilis* has reported decrease in expression of several biofilm forming associated proteins (*comA, epsK, epsK, tuaD, epsC* & *tuaD*) (Pandit et al., 2017). Furthermore, RT-PCR analysis of sodium ascorbate-treated *Pseudomonas* sp. has confirmed the reduction of signaling molecules C4-HSL and 3-oxo-C12-HSL by repressing the expression of Quorum sensing regulating genes such as *lasI, lasR, rhII, pqsA*, and *pqsQ* at molecular level (El-Mowafy et al., 2014).

3.8. Effect of VitC coating on growth and survival of EMC 17 in paneer food matrix

Evaluation of the present investigation in preventing the growth of biofilm forming *E. coli* was performed *in situ* via testing the efficacy of VitC coating on Indian cottage cheese (Paneer). Meanwhile it is important to mention here that VitC acts as an edible coating because of its non-toxic effects. Therefore, experiments were designed and performed accordingly. We chose Paneer because of its huge indigenous production in India as well as it being a leading dairy product with a very low shelf life (~1 day at room temperature and 6 days at 10 °C) (Kumar et al., 2011). So far, many techniques viz. low temperature, high relative humidity and modified atmospheric packaging have been employed for extending the shelf life of paneer (Mutaku, Erku, & Ashenafi, 2005; Sharma et al., 2019; Tsegaye & Ashenafi, 2005). However coating methods that are edible in nature not only enhances the quality of the food products by preventing the physical and chemical deterioration but also help in evading microbes in a cost-effective manner. Sharma et al., (2019) have recently reported the application of various natural compounds (phytochemicals) along with other hurdles

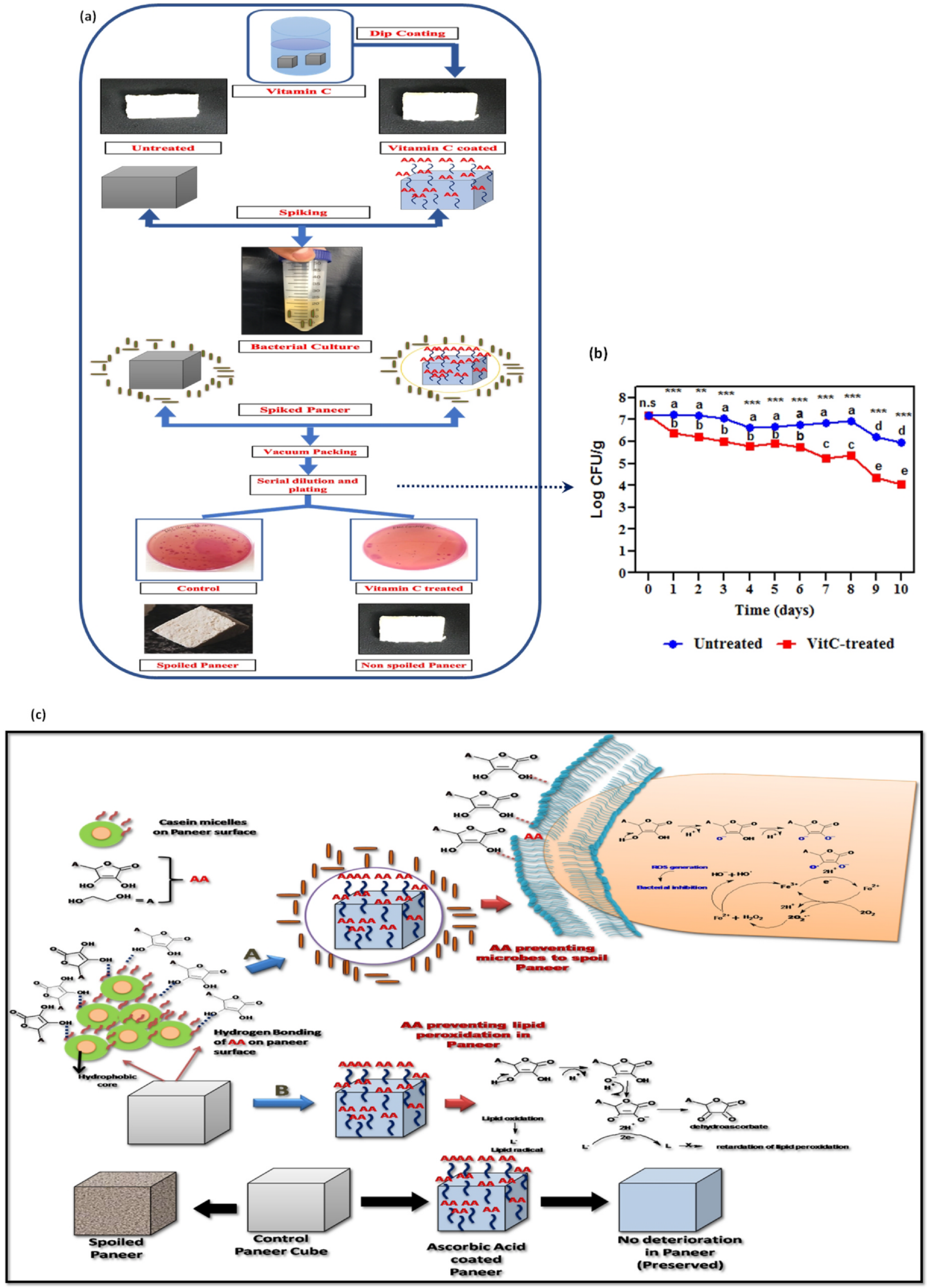

Fig. 4. *In situ* application of Vitamin C as an antimicrobial coating of Paneer (Indian cottage cheese) (a) Experimental set up ensued to test the *in situ* efficacy of vitamin C coating in controlling growth of *E. coli* EMC17 isolate over a period of time; (b) Abrogation of *E. coli* EMC17 isolate survival within Paneer over a period of 1–10 days at 4 °C. (c) Proposed mechanism of action for paneer preservation depicting (A) antimicrobial coating of vitamin C as prooxidant (B) antioxidant behavior preventing lipid peroxidation; Mean ± SD from 3 replicates plotted for panel c; Means with different letters are significantly different (p- value *** < 0.001, ** < 0.01, ns, not significant).

as promising treatment in controlling the growth and survival of *E. coli* in paneer.

The *E. coli* EMC17 count was evaluated on paneer, i.e dip coated with VitC that was vacuum packed and stored at 4 °C. Since it is a well-known fact that the survival of bacteria (*E. coli*) at refrigeration temperature under *in vitro* and *in situ* conditions, reflects its adaptability to survive in extreme conditions (Mutaku et al., 2005; Sharma et al., 2019; Tsegaye & Ashenafi, 2005).

The results in Fig. 4b showed significant reduction in *E. coli* count by ~ 1.5 logs (4.03 log count) after 10th day of storage as compared to untreated control (5.97 logs). The currently undertaken *in situ* study within perishable foods like paneer undoubtedly/unambiguously establishes the efficacy of VitC as an antimicrobial that brings about reduction in the growth and survival of EMC17 within food. In fact, VitC is playing a dual role as a shelf life extender of paneer by acting as a pro-oxidant (mentioned in mechanism section 3.5) for evading microbes (Fig. 4c-A) and also prevents the lipid peroxidation (Fig. 4c-B) by acting as an antioxidant. Briefly the Fig. 4, depicts the plausible mechanism of the VitC action as an antimicrobial coating. Paneer, which is a cluster of casein micelles is majorly Calcium phosphate; the inner core of the casein micelle is hydrophobic, and the outer core is hydrophilic. The hydrophilic chains present on the top layer of the Paneer interact with VitC (Ascorbic Acid, AA) via H-bonding. Once AA coating is performed on Paneer (via repetitive dip coating method), followed by bacterial invasion (during bacterial spiking), AA clings to the bacterial membrane (polar interactions) and this allows penetration of AA inside the bacterial cell. Once AA enters the cell it inhibits the functioning of bacteria while forming ascorbate ion (which acts as a pro-oxidant releasing protons); this initiates the ROS generation via fenton's reaction and ultimately leads to the oxidative damage, DNA denaturation and bacterial killing.

In addition to the significant decrease in the microbial count, the physical examination and sensory evaluation of the dip coated paneer, tasted good showing the presence of no off-flavours and odors in paneer (Fig. 5). While the unpleasant flavour and off taste in control paneer, implicated to the lipid peroxidation which has been explained mechanistically in Fig. 4b where in VitC (AA) being an antioxidant quenches the formation of lipid radical by accepting the radical by ascorbate ion to form dehydroascorbate. The mechanism clearly depicts the pro-oxidant and antioxidant behavior of VitC as an edible coating on paneer and this can be a resourceful methodology in preserving paneer at a large-scale level.

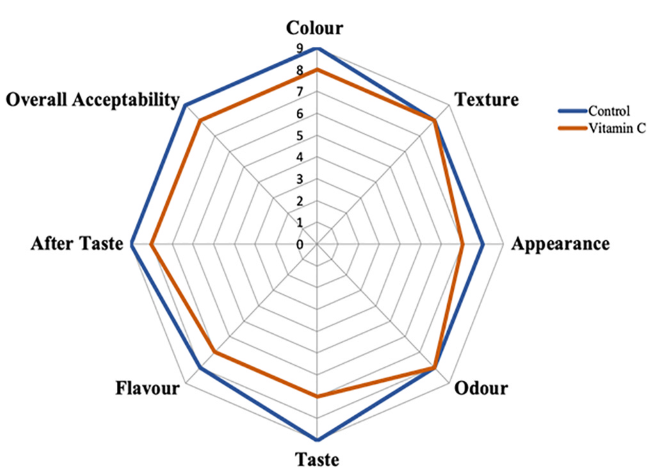

Fig. 5. Sensory Analysis of vitamin C coated Paneer: Spider chart representing mean score of the evaluated sensory attributes of experimental paneer sample treated with vitamin C and untreated control.

3.9. Sensory analysis

Fig. 5 reports the sensory analysis of the prepared paneer samples. The VitC coated paneer sample was found to be accepted by all panelist in terms of appearance, taste and flavour except few who reported slightly sour after taste.

4. Conclusion

The escalating threat of antibiotic resistance amongst pathogenic bacteria and toxicity of available drugs demands for urgent innovative approaches wherein natural and safe antimicrobials with distinctive mechanisms and broad antimicrobial spectra are made available for applications in various industries including healthcare and food industry. In the present study, the efficacious application of pro-oxidant VitC to serve as a natural and eco-friendly, *in situ* antimicrobial intervention against multi-drug resistant, biofilm-forming *E. coli* and possibly other gram-negative pathogens which pose a formidable challenge for antimicrobial therapies was demonstrated. By virtue of its pro-oxidant behaviour at increased concentrations, VitC generated an oxidative stress by production of ROS thus resulting in loss of cell viability and impairment in metabolic activities; cell signaling (quorum sensing), EPS production, leakage of intracellular proteins, sugars and down regulating biofilm gene expression of pathogenic *E. coli* EMC 17. In view of the above VitC was employed as an edible coating on paneer resulting in disruption of the microbial cellular homeostasis and viability thereby preventing the microbial invasion and growth on food surfaces. This has a huge potential application as a natural food preservative and anti-biofilm therapeutic with immense scope of revolutionizing the area of microbiological food safety. Finally, this study is the first to provide evidence for the mode of action of VitC against MDR pathogenic *E. coli* isolated from meat samples.

5. Author contribution statement

NKT contributed in conceptualization of the project, methodology designing, data analysis, and overall supervision of the study. SDP and AL contributed in data curation and data analysis. NKT, SDP and DS wrote and revised the manuscript. All authors reviewed and approved the final manuscript.

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Ethical approval

All procedures performed in studies involving human participants were in accordance with ethical standards of the institutional and/or national research committee. Informed consent was obtained from all individual participants included in the study.

The authors have no conflicting interest.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2020.128171.

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